



Computational Physics

Intro

Molecular dynamics simulation of Argon

Monte Carlo simulation

Project description

Lecture notes

Basics of the Monte Carlo method

Intro: Monte Carlo simulation of the Ising model

[Intro: Variational quantum Monte Carlo](#)

Intro: Monte Carlo simulation of polymers

Final, free project

Computing crash course

Project and programming skills

Variational Quantum Monte Carlo

The stationary Schrödinger equation can only be solved exactly for some specific examples. In general, and certainly for interacting systems, an exact solution is impossible. Still, we can try to find the approximate ground state energy using a variational calculation.

To this end, we consider a *trial wave function* $\psi_T(\mathbf{r}, \alpha)$, where α is one or more variational parameters and $\mathbf{r}$ denotes the spatial degrees of freedom. We can compute the expectation value of the energy as

$$E(\alpha) = \frac{\int d\mathbf{r} \psi_T^*(\mathbf{r}, \alpha) H \psi_T(\mathbf{r}, \alpha)}{\int d\mathbf{r} \psi_T^*(\mathbf{r}, \alpha) \psi_T(\mathbf{r}, \alpha)}$$

The variational principle says that $E(\alpha) \geq$ the real ground state energy. The best approximation to the ground state energy is thus obtained by minimizing $E(\alpha)$ with respect to α . Obviously, the quality of the approximation depends on how good the trial wave function is.

The Monte Carlo method is in this case used to compute the integral over the spatial degrees of freedom represented by $\mathbf{r}$. For the case of the helium atom, these are 6 degrees of freedom (positions of 2 electrons in 3D space), and this is high-dimensional enough for the Monte Carlo method to be useful.

To get a proper probability, we define the local energy as

$$E_{\text{loc}}(\mathbf{r}, \alpha) = \frac{H \psi_T(\mathbf{r}, \alpha)}{\psi_T(\mathbf{r}, \alpha)}$$

Then the energy can be computed as

$$E(\alpha) = \frac{\int d\mathbf{r} \psi_T^*(\mathbf{r}, \alpha) \psi_T(\mathbf{r}, \alpha) E_{\text{loc}}(\mathbf{r}, \alpha)}{\int d\mathbf{r} \psi_T^*(\mathbf{r}, \alpha) \psi_T(\mathbf{r}, \alpha)}$$

and we can solve this integral using Monte Carlo integration with importance sampling according to the probability density $\psi_T^* \psi_T / \int d\mathbf{r} \psi_T^* \psi_T$ - which we can again sample exactly using the [Metropolis algorithm](#). The configuration space we sample then is given by the position $\mathbf{r}$ (or the positions $\mathbf{r}_1$ and $\mathbf{r}_2$ for the Helium atom). A Monte Carlo move then consists of choosing a new position $\mathbf{r}'$ randomly from a region around the old position. Make sure you do this such that the probability of going forth and back is the same, as required by detailed balance (this is achieved most easily by using Cartesian coordinates, and choosing rectangular regions). Also, make sure to choose a region of a size such that the acceptance ratio is not too big or too small.

As a starting point, apply the variational Monte Carlo method to a solvable model, such as a harmonic oscillator or the hydrogen atom. You will need to

- derive the expression for the local energy for a given trial wave function
- write a code that can implement the Monte Carlo integration. Make sure you make a move such that the overall acceptance ratio is not too low or too high.
- minimize the energy with respect to α . This is a bit more subtle as it seems, as the Monte Carlo result for $E(\alpha)$ has an error associated with it, and the usual minimization algorithms may be thrown off track. Care must be thus taken to compute the gradient.

After that, solve the problem for the helium atom for which no exact analytical solution is given.

This project is thus rather predefined and follows a fixed path. There is less freedom than in the other applications mentioned above, which are more difficult to implement. Yet, if you come up with an idea, feel free to discuss with us!

Milestones

In the second project, we only provide you high-level milestones for every week! You will need to define every week your own detailed milestones.

- **Week 1:** Start to implement a basic simulation to compute the expectation value of the energy for a given set of variational parameters, using the Metropolis algorithm. Argue why your approach fulfils detailed balance and make plans how to validate your code.
- **Week 2:** Finish the basic simulation, monitor the acceptance ratio and validate your results.
- **Week 3:** Generate results for the Helium atom (including error bars) and start to implement the proper minimization of the energy.

Literature

- details on the variational method can be found in [Chapter 4 of the lecture notes of Applications of Quantum Mechanics](#)
- implementation details of the Monte Carlo method in [Chapter 12.2](#) from the book "Computational Physics" by Jos Thijssen.

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